2025-06-11 NSF S&CC – CommonSenses project – sensor box data specifications

Multiple types of messages pushed up through the Particle interface

Type #1: data from a sensor box

We collect temperature (T), relative humidity (RH), and noise sensor readings every minute. However data are only pushed up to the Particle.io system every 15 minutes (this saves power in booting up the LTE module only infrequently rather than keeping it open).

This means one "message" will have 15 T, RH, and Noise data readings, each for a consecutive minute.

The message starts with the timestamp ('month', 'day', 'hour', and 'minute') as **UTC time stamp** of the first sensor reading in the list). For each subsequent reading we should add 1 minute to the time stamp.

The sensor data are also uploaded as integers that need post-processing to get to the units we want to report in, also to minimize data transmission (power consumption).

Generic format of a message:

,BoxID,month,day,hour,minute,T,RH,Noise,T,RH,Noise,T,RH,Noise,T,RH,Noise,T,RH,Noise,T,RH,Noise,T,RH,Noise,T,RH,Noise,T,RH,Noise,T,RH,Noise,T,RH,Noise,T,RH,Noise,T,RH,Noise,T,RH,Noise,T,RH,Noise

Example:

,19,6,6,17,34,2483,5480,6389,2482,5483,5181,2477,5481,5222,2484,5486,6121,2485,5487,5649,2483,5481,6591,2483,5482,6923,2488,5484,6904,2484,5482,6860,2480,5482,6731,2487,5488,6829,2487,5491,6762,2485,5486,6515,2479,5488,6711,2488,5489,6806

In the example:

- Box ID is 19
- Time stamp (of first sample) is 6/6/2025 17:34 UTC time – i.e., the first T, RH, and Noise data were collected at 17:34 UTC time, and the second T, RH, and Noise data were collected at 17:35 UTC time, etc.
- The first "batch" of data are: T_raw = 2483, RH_raw = 5480, Noise_raw = 6389

Postprocessing needed:

- Temperature should be **divided by 100** to get to the actual reading **in degrees C.**
    - In the example above T_raw = 2483 → T_actual = **24.83 C**
    - The sensor only has accuracy around 0.1 C at best, so this can be rounded to keep just one decimal of precision in our database.
- Relative humidity should be **divided by 100** to get to the actual reading **in %.**
    - In the example above RH_raw = 5480 → RH_actual = **54.80 %**
    - The sensor only has accuracy around 1 % (RH is kind of tough to measure), so we can choose to round or keep one decimal of precision in our database.
- Noise is a count-based reading from a 16-bit analog-to-digital converter (signed integer) and must be converted to **dB** using the following equation: **dB value = (Noise_raw / 32768) * 6.144 * 50**
    - In the example above Noise_raw = 6389 → Noise_actual = (6389 / 32768) * 6.144 * 50 = **59.90 dB**
    - Depending how we want to treat analyses, we might want to keep 1 digit of precision, but for displaying people probably don't need to see anything other than the integer part of the number.

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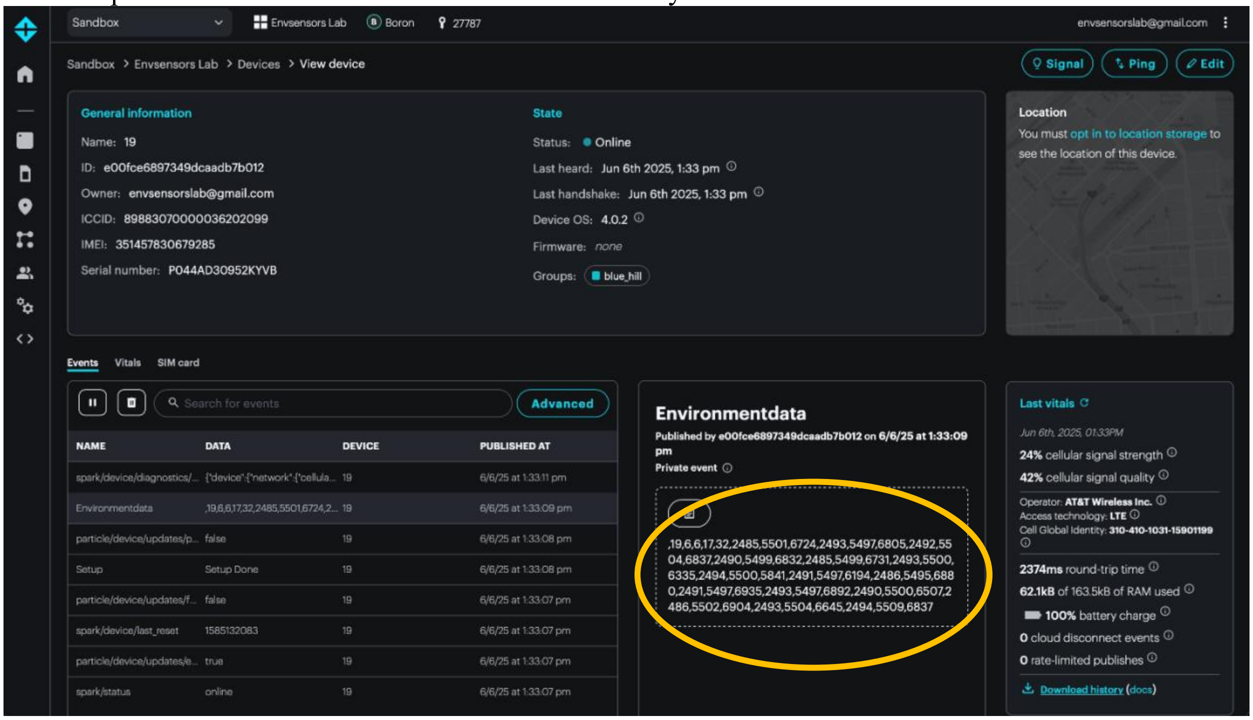
An example of how this will look like in the Particle.io system is here:



Type #2: error reported from a sensor box

Errors that happen in the code which will indicate a hardware malfunction are also reported. These would ideally be reported via email or similar to the research team, as it would indicate some physical action (probably visiting the sensor) is needed.

Generic format of a message:

BoxIDplusDate Hour:Minute ErrorCode

where *BoxIDplusDate* is 2-digit BoxID followed by 2-digit year, 2-digit month, and 2-digit date

Example (first highlighted message in screenshot below):

19250610 14:42 E10

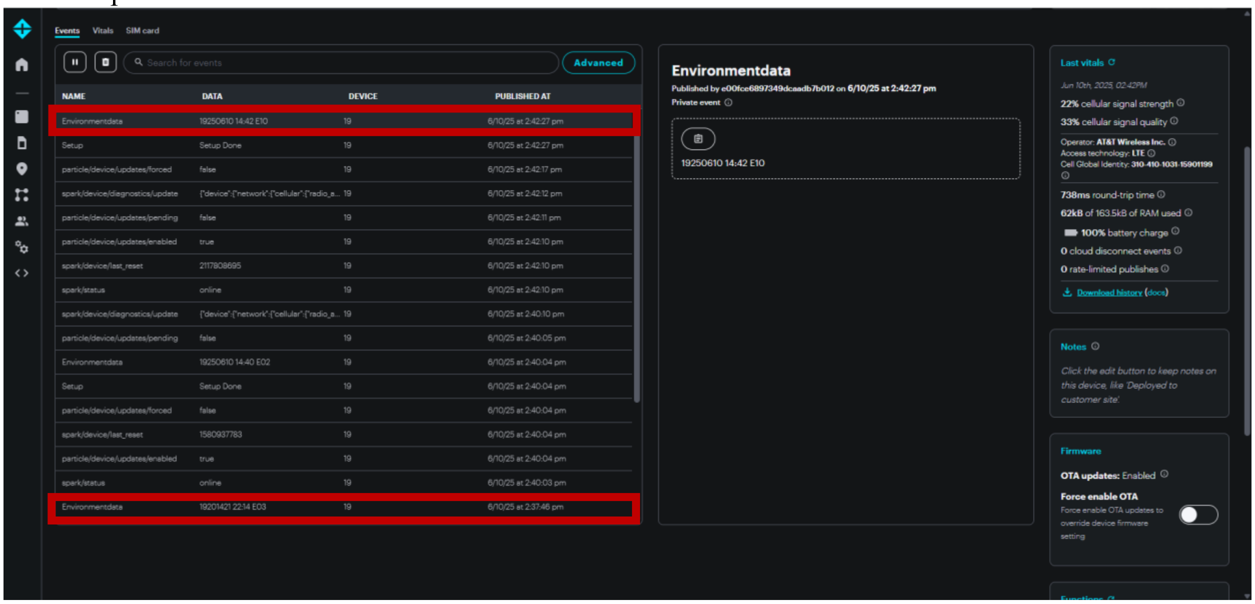
In the example:

- Box ID is 19
- Date and time: 2025-06-10 14:42 UTC
- Error number is 10 (from table below this means "ADC not found")

Error Code	Error Description
E1	Serial Port not open
E2	T&RH sensor not found
E3	RTC not found
E4	LTE module connection to cloud timeout (only log to SD card)
E5	LTE module receiving data from Arduino timeout (only log to SD card)

E6	SD module failure
E7	Buffer overflow
E8	SD open file failure
E9	SD Error logging failure
E10	ADC not found

An example of how this will look like in the Particle.io interface here:



Note: the last highlighted message has wrong time stamp because the RTC is missing at this time (E03 means RTC missing).

Type #3: startup message

Every time we reboot the whole system (at startup or if the Arduino is restarting after losing power), the Arduino will send a "wakeup" message to Particle.io.

If we get the following message on the Particle.io, that shows the success of the connection of the cellular module to internet and the communication between the Arduino microcontroller (box "brain") and the Particle.io cloud system.

Uses:

1. When we are installing sensors, this is the "quick" message that will allow us to verify the sensor box is functioning properly.
2. If we see this message at any other time, it is telling us the box lost power. While it is possible that is just a reality in winter, **at any other time this would probably indicate some kind of malfunction of the battery or the microcontroller.**

Generic format of a message:

BoxIDplusDate Hour:Minute **LTE Setup Done**

where *BoxIDplusDate* is 2-digit BoxID followed by 2-digit year, 2-digit month, and 2-digit date and

where "LTE Setup Done" is text that is always the same (a static wakeup message).

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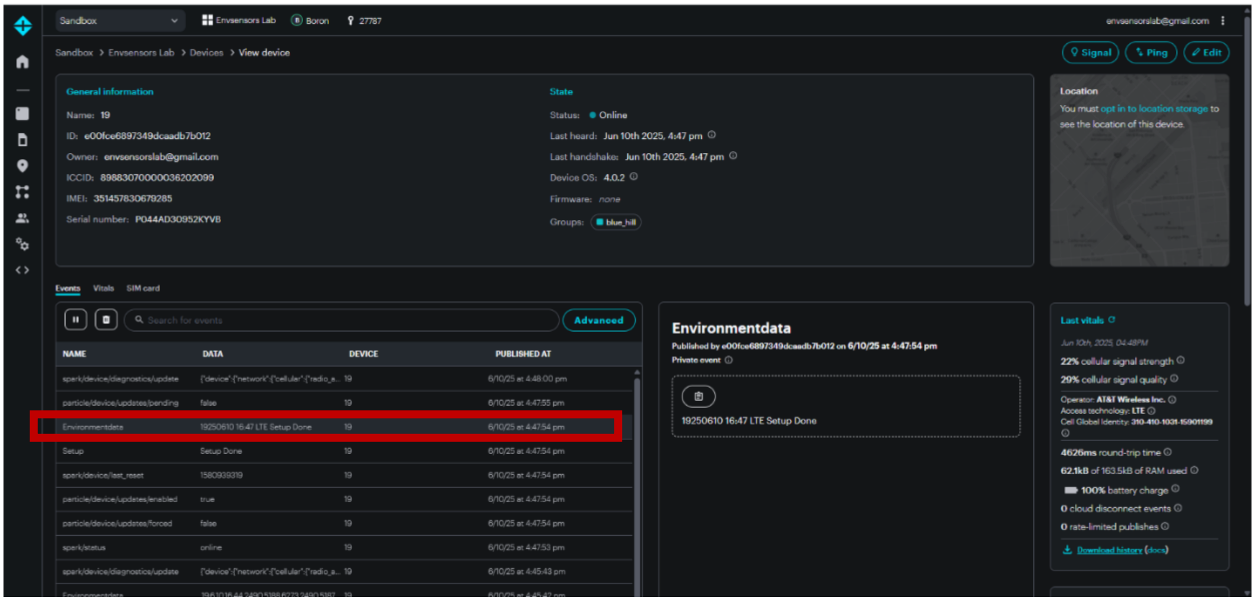
Example (in screenshot below):
    19250610 16:47 LTE Setup Done

In the example:
- Box ID is 19
- Date and time: 2025-06-10 16:47 UTC
- Box is reporting bootup and successful connection to LTE network



__Type #4: system information__
Note that the Particle.io system also provides other helpful information about each box, including the signal strength, last connection, etc. **These could be useful for diagnostics, and we may wish to consider pulling these at least once/day with each data message as sensor health checks**.